

# 11-12 Performance Review

## Manager Review

Jonathan Daniel Micklos

Performance Review: 04/01/2011 - 03/31/2012

**Reviewer:** Michael Nail

**Role:** Manager

**Status:** Completed

**Manager:** Michael Nail

### Overall Performance Summary

#### Employee Summary:

This has been a very big year for me. I have not only switched teams, but was a part of the Kindle Fire, Kindle Touch and latest generation Kindle detail page launches and ship dates. In addition to being a part of massive undertakings such as these, I have continued to work on projects of smaller scope, continued my work mentoring and proactively begun work on things such as build and CSC obfuscation tools, all the while continuing to show that I am turning my areas for improvement in my past team (Earn Trust of Others and Have Backbone; Disagree and Commit) into strengths and in doing so continuing to perform safely in the SDE II level.

The two main projects that I was a part of since joining the KiSIT team were the TWO detail page launches and the TWO first-ship events. For the TWO detail page launch, I was responsible for designing, implementing and testing the Kindle Family Stripe. For which, I designed it to be highly configurable and robust. Designing it to be highly configurable significantly reduced the amount of developer time needed alterations to the Kindle Family Stripe. Removing the need for code changes for future additions/subtractions/modifications of Kindle Families in the stripe has been very valuable.

In addition to the Kindle Family Stripe, I took up the task of re-writing the Image Block for the new Kindle Device detail pages with just enough time to spare for appropriate QA work and iteration. The initial Image Block that was to be use in the TWO detail pages was having bugs being repeatedly solved and reemerging. After some investigatory work on my own, I had seen that the bugs were both hard to reproduce consistently and were systemic to the overall design of the Image Block. In the interest of making sure our customers had the best user experience possible, I worked to re-implement the Image Block free of these bugs.

Apart from these two core components of the detail page for the TWO Launch, I also worked to configure multiple different features alongside others on the KiSIT team. This included tasks such as creating obfuscated content for the new pages and tracking and resolving ongoing issues with individual features. Because our bug list was somewhat varied in both priority and type, I took it upon myself to organize all of the incoming bugs into a wiki so that they could be communally triaged and fixed by our team. This wiki setup, provided a much more agile method for tracking issues that would be traditionally housed in JIRAs or TTs and provided a one stop center for gauging the current health of the TWO detail pages in preparation for launch.

For TWO Ship, I worked on two main projects. First, I re-wrote and re-factored much of the Kindle CSC mason code to support the Kindle Fire. This was a large undertaking as not only does the Kindle Fire introduce the new paradigm of multiple device types associated with a single device (a

device type for the device and then for each of the software components), but also from inter team organization perspective. While I was making this code change, the CSC team was in the middle of refactoring much of the Kindle CSC code base.

For TWO Ship there was also an emergent need for the ability to send targeted Kindle Device Letters (PSNLs) to customers who had yet to update their Kindle Fire firmware. For this I wrote a system scans for such customers on an hourly basis and sends PSNL send requests for each to the Kindle Letter Service. I designed a system that was able to accomplish 65 tps (PSNLs sent per second) for peak traffic. I presented this design in a technical review and made sure to add the ability to throttle the system.

In addition to these projects, I continued my work on continue to raise the bar at Amazon from a personnel level. I officially mentored two interns and continued to mentor others in an unofficial capacity. Additionally, I continued to participate in interview loops and hiring meetings.

As I said early, this has been a very big year for me. Joining the KiSIT team has opened a world of new opportunities for me in which I have been delighted to be able to help further Kindle and continue to support that Amazon is earths most customer centric company.

### **Manager Summary:**

Jon showed great commitment and determination overcoming some personal performance issues during the first quarter of the year. Jon was placed on a performance improvement plan from February to March in order to address occasional displays of anger/frustration (both within the team environment and in written Remedy ticket correspondence) and inconsistent focus on assigned sprint project work. Despite some initial reluctance about being placed on the PIP, Jon focused, addressed the issues and successfully completed the PIP.

From that point on Jon rededicated himself to being a strong teammate, actively driving his sprint project work and continually exuding a positive attitude. He refactored previously developed features like image hover/zoom functionality and immersive view. His work here simplified the design and code base as well as decreasing the client side latency related to the original implementation. Even though Jon did not feel as technically challenged as he wanted to be on these projects, he understood the need to complete them. From there he enhanced the BuyBox and Order Pipeline (BBOP) detail page feature we shared with the Prime team, adding functionality that allowed editing of text strings and display of related metrics within Content Symphony. He took on a late breaking firedrill project (enabling a free shipping promotion for FR) since he had the available bandwidth at the time (other team members were heads-down trying to finish sprint commitments). He mentored intern Mattia Padovani on creating a new widget (Have One To Sell) which we tested in Media categories. And he also gave consultation to external teams like the ICE team, helping them work through dependencies for the Spain website launch.

Jon was also very active in our operational duties: code reviews (internal and external), on call shifts, and providing external consultation (e.g.office hours, one-off meeting requests from external teams).

Jon's efforts to address his performance issues, coupled with his strong performance and dedication after addressing those issues, helped the Buying Experience team immensely.

Jon is very passionate, both on a personal and work level. That passion drives his strong bias for action. When he harnesses that passion effectively he provides strong technical leadership and delivers results. That same passion sometimes got the best of Jon in late 2010 and into early 2011. He voiced displeasure in ways that some internal and external team members found inappropriate and made them uncomfortable. Jon addressed those issues during his PIP and then performed very strongly during the remainder of his time in Buying Experience. I'd recommend he remain vigilant about keeping a positive attitude and not reacting too quickly in situations where a more measured

response can be more effective.

Jon quickly wrapped up his performance plan and joined the Kindle team shortly thereafter. He has been enthusiastic and a pleasure to work with by all accounts. He has been a role model for the engineers on the team, giving guidance to the various projects we have taken on. He has been instrumental in the success of device launches, letters, Family Stripe, immersive view, and many other projects.

## Identify Strengths

### Employee Summary:

#### Bias for Action and Deliver Results:

I value my bias for action. As a developer it is a very integral part of who I am. When it is apparent to me that action is necessary in order for a project to meet reasonable deadlines, I thrive on being able to step in to make it ultimately succeed.

One example of this occurred right before the TWO launch. About two weeks prior to the TWO launch, the new Image Block (which was being developed by our team) was not functioning correctly. During QA testing, bugs were continually being fixed and reappearing. I was concerned in that the new Image Block would be launched with a significantly lower quality bar than we would otherwise be ok with and that further cycles on the same bugs would starve other features of needed attention from SDEs.

I did some research into the bugs and what could be causing them. I found the issues to be both somewhat inconsistently reproducible and systemic of the overall design of the widget. At first, I had suggested that the broken functionality be delayed for further triage and launch until after the TWO launch. After finding out that similar functionality had seen significant positive weblab results on other device detail pages, another option was needed.

In order to make sure that the new functionality made it into the TWO launch, I re-wrote the Image Block widget from scratch over the course of two work days and an all-nighter. The code was submitted and passed QA in time for the TWO launch. No significant bugs have been found in production with the new Image Block.

Another example of my Bias for Action can be found in a project that was never launched in production. At the time, it was thought that there would need to be a way to send targeted PSNLs (Kindle Device Letters) to customers who had not updated the firmware on their Kindle Fire. The need for such functionality was presented shortly before Kindle Fires would begin shipping. In the matter of a few days, I was able to design a system that would hourly check for customers who have yet to update the firmware in an hour to two hours since they initially turned the device on and connected to the internet. The customer devices that were on this list would then be sent PSNLs explaining how and why to upgrade. Shortly followed by a couple days of work, I was able have a working example that was both throttleable and could achieve a maximum of 65tps (PSNLs enqueued to be sent) in order to more than satisfy peak traffic estimates.

#### Insist on the Highest Standards and Customer Obsession:

High quality is of the utmost importance to me. Bugs and poor user experiences greatly lower customer expectations and trust. As a customer of Amazon.com, I have very high standards of how I should be treated as a customer. As an SDE at Amazon, I know that nothing less would suffice.

Meeting such a bar is not easy. This is especially true when writing customer facing website code.

Because there are often many permutations of different environment settings that a user may have for accessing the web, it is very important to be thorough to test each. To accomplish this, when making UI altering code changes, I design a test matrix of user environments to test. In this test matrix, I consider a number of different criteria (depending on the specific code change). Different browsers, different browser versions and JavaScript Enabled or Not are just some of these criteria. In code reviews I make sure that this test matrix, alongside clear testing instructions is included. This allows me to communicate to code reviewers as well as QA not only what I have tested but also how to reproduce the steps. Any further recommendations for testing edge cases can then be made.

Being this thorough has led to no major issues with customer facing code I have written while on the KiSIT team. Additionally (as pointed out by Steven Miller to me), my code review style that I use, has been adopted by other members of the team.

### Ownership:

Having ownership is core to not only Amazon, but also good software engineering fundamentals. Good code is code that is written to be reusable both now and well into the future. Good code is code that satisfies current use cases and prudently anticipates those in the future. I write code as an owner.

One example of this is the code I wrote for the Kindle Family Stripe. As pointed out in my goals, I designed the Kindle Family Stripe to be highly configurable by our business owners and copy editors. Building the Kindle Family Stripe to be this configurable has empowered our business owners to make changes to both drive sales and respond to changes in device availability without code changes. Examples of this include the modification (and then removal) of the Kindle Keyboard from the family stripe in response to availability and the addition of the Kindle DX in order to drive sales.

Also as an owner, I do not say "that's not my job." One example of this was previously mentioned was the work I did to get the Image Block working.

### Manager Summary:

Jon exemplifies many of Amazon's leadership principles. His combination of high standards, great technical skills, and positive attitude makes him a role model for his peers. His can-do attitude has lead him to get involved in all aspects of software that we build from front-end to back-end.

He always thinks of the customer experience and impact first, and thinks big about what he is designing and building - for example, when thinking about device launches he was quick to the approach of consolidating knowledge and automating where possible. When a late requirement arrived to build a website widget representing the Kindle family, Jon was able to provide ample guidance to get it launched on time, designed such that it has been adaptable to multiple changes since with no further engineering effort.

"He has proven to be motivated and hard working. Jon loves to drive a project and its requirements: for example, he started working on product bundles for the detail page, when it was essentially just the beginning of an idea. He quickly identified blockers and other issues that would need to be resolved before more work could be started. This was extremely useful so that our team could set reasonable deadlines for the project."

"Jon also started work on a self-service version of the detail bullets on the detail page. This project was aimed to prevent the need for developer intervention for small requests and additions to the feature. He worked with our product managers to define the requirements and developed a design for the project. He left our team before he could do much more with the project; however, he was able to hand it off smoothly, so it could be continued in the future. "

"Jon showed this past year that he is willing to dive deep and work on a variety of different tasks that help our customers and improve our code base, in general. Jon was the lead developer for improving the customer experience and changing the Prime sign-up button in the buy box to a check box, working on adding additional reftags for our features, adding quantity restriction messaging to the buy box, and deprecating our DE low price guarantee code, as well as the CYOR (create your own ring) code paths. For some of these projects, he worked with an intern on our team, and he displayed a strong aptitude for mentoring and developing a teammate. Jon also worked on internationalizing the immersive view and hover zoom features for the image block to all locales. While doing this, he had to dig deep into the code to find latency improvements, so that the features could be launched."

"Jonathan started re-writing stretchy goodness feature just a week before T/W/O Launch. He built a working version of stretchy goodness independently and integrated with AmazonFamilyKindleDeviceMason package. He was willing put in extra hours to meet the deadlines. The team was able to address all the launch release blockers associated with stretchy goodness"

"Jonathan is very passionate and dedicated in doing the right thing and this shows up in his ownership of development projects like the Family Stripe and the "Stretchy Goodness" main image project. I appreciate how open he is to feedback and willingness to go the extra mile to make sure things are right. The amount of dev testing he does is excellent and the documentation he writes for QA on what has changed and how to test it is exceptionally good."

"I also appreciated how when he took on the role of owning the builds for the Device Sales maintenance release that he created a set of scripts to automate the build and deploy to devo and gamma process. These scripts not only decrease the amount of time needed building each release but also increases the quality of the releases because it decreases the places where human error may occur. He also did a great job of involving QA during his planning so that our needs are also met."

"Jonathan displayed ownership when he worked on the CSC for Kindle Fire. There were no clear requirements that were documented and he took the initiative to collect requirements from the Kindle CS Program Managers and develop against a set of requirements. He also conducted a demo of his work with the CS Operations and PM to ensure that there weren't any missed use cases."

"During my first on-call period, when I knew absolutely nothing about the code base, I was paged for a nasty non-secure inclusions issue on a Saturday afternoon. Jon came into the office for 4 hours and helped me learn how to debug issues with the detail page, and helped me nail down the offending component. I really appreciated his willingness to go above and beyond to help me out while I was still new."

"I am very impressed for Jon's bias for action. When we were working together with the 2011 new device launch, there were several blocking issues with the product image block. Jon rolled up the sleeve immediately and started a better redesign of the image widget. The rework is great and there is no issue found for the new widget.

Jon also demonstrated good ownership. After he got the ownership of building/deploying release candidate package to pre-prod hosts. He created a new tool to handle the deployment which addresses several issues with the original Gemini tool.

Jon also did a great job when doing phone screen or on site interviews. He always provides precise and detailed feedback for candidates. He has set good examples for other SDEs to follow and he definitely raised the bar for hiring good candidates."

"I worked w/ Jonathan briefly on (1) customer service tool changes related to Kindle Fire and (2) planning for an update letter on the Kindle Fire. I saw that he quickly pointed out some long-standing issues w/ the customer service tool design and came up w/ ways to fix for the future. And although (2) never came to fruition, his design review was thorough and well thought through."

"Jon was a very helpful "BX teacher". He spent quite some time explaining me how the systems work, from development to release process, writing long and detailed emails for me to have a clear picture. He was also very attentive at the quality of the code reviews he did for me, insisting on understanding the change and testing for all the edge cases."

"Jon is a man of passion and is trustworthy. Real dedication can be seen in tasks he takes up and if that is also something he likes from heart he would \*do whatever it takes\* to get it done. It was amazing working with him while he was in BX team. He did awesome job with test-driven-development approach on Books expansion project he delivered, from which I learned a lot."

"Set the standard for RMD line item documentation and Code Reviews with Testing Data. The team should review Jonathan format when submitting RMD line items and Code Reviews."

"Jonathan is hard working. I have witnessed that he spent many extra hours on finishing important tasks, in which the stretchy goodness task was a good example. At that time, the issue in stretchy goodness was a big blocker to our launch, and Jonathan volunteered to take on this task, and delivered a high-quality product.

Jonathan is willing to help and share knowledge with others. I benefitted a lot from his help when I first joined Amazon. He spent time sitting with me on troubleshooting the problems that I had about the Apollo environments. There was another time when I started to work on the UI part of Immersive View. He also helped me break down the task and helped me with the time estimates, which gave me a good start.

He is responsible for the code review requests. He provided a lot of good suggestions for improving the software. Through him I gained a lot of good insights and valuable feedback from him. He also pointed out some potential risks from the code. For example, when I was requesting code review for the Family Page task, he pointed out that I quoted the absolute paths of image assets in CSS, which posed a security issue. He also tested the code thoroughly on all the browsers to make sure the page works. I could see that he presented his co-ownership of the code as a reviewer and kept the high standard of code quality.

Jonathan has demonstrated a key ability to craft creative solutions to problems. He showed me the script he created for building and deployment, which made the process a lot easier.

Jonathan is very good at communication as evidenced by his effectiveness in seeking clarity on requirements and understanding between product managers and engineers. There were times when I was not clear, he advised and also led me to talk to PMs to communicate with them. I have learned a lot from him."

"Jon is fantastic at taking charge and just getting things done. I was pretty impressed by the way he just rewrote the image block after Stretchy Badness bugs kept dragging on, and on, and on. He also is super knowledgeable and helpful, and always willing to teach (not just solve someone's problem for them, which is an important distinction.) He also puts a lot of effort into writing \*good\* code as well as testing it very thoroughly. If you give him something, you can be sure that it will get done, and done well."

"He has worked extensively and tirelessly with the CSC support for the Kindle Fire. And when there was a major error, he is willing to stay late at night to fix the problem. His dedication exemplifies customer obsession and ownership."

"He is also a very good motivator. He often acknowledges people's work and effort and is always willing to help others facing problems. For example, he was willing to explain to me how Twister works and his help was monumental to me and my project for international DTCP."

"Jon has done a good job integrating into the team and taking ownership on projects. He has a high standard and is able to instill that mindset onto others. When working on code and learning about a new system, he thinks big and tries to improve things rather than maintaining the status quo. An example of this is for the Kindle family stripe or Kindle fire CSC projects, where Jon actively worked to make the code base better. In the case of the family stripe, as it was a new project, he worked hard on making it flexible and adaptable to changing situations."

## Identify Areas for Improvement

### Employee Summary:

#### Frugality:

Looking back at the time I have spent on the KiSIT team so far, I know that one of the reasons our team is so nimble is with the outside help from other teams. Because we are working on the Kindle, our team is afforded a lot of power to call upon the help of other teams at a moment's notice, in both the interest of keeping to launch schedules, but also to prevent leaks. As a team, we must be careful to use this capability with discretion, not just because it is available. Especially as someone who has worked on other teams in Amazon who has been on the opposite side of the conversation, I know that getting a reasonable amount of heads up about need for another team's involvement can be incredibly helpful. Going forward, when in need of help from other teams, I will work to maximize the amount of heads up time I give when is possible and/or reasonable.

#### Think Big:

On technical matters, I generally take pride in thinking big. However, thinking big should not reside as a tenant for coding and systems design alone. While continuing to ask "What is the technical impact that will be made?" I want to also ask "What is the human impact that will be made?" There are two ways that I will work on this approach going forward:

Reduce Required Human Involvement for Expansions: In the three and a half years I have worked at Amazon, the large-scale projects I have worked on could generally be classified as one of the following:

- Projects which directly improve the customer shopping experience through new and/or optimized user interfaces
- Projects which reduce developer pain by refactoring unwieldy code libraries
- Projects which expand a given user experience to new categories or locales

In the last example, expansion projects (example: twisterizing a category in a given locale) are possible because of a strong foundation of robust tools, repeatable steps and thorough testing. With better tools and simpler steps, less human involvement is required. In line with thinking such as that behind Hammer Time, I wish to design automations, workflows and processes which are robust as to allow for the quick expansion of given Amazon offerings into new locales and categories (ultimately, reducing the need for human involvement; allowing for manual innovation over manual expansion).

Inspire Others to Think Big Organically On their Own and as a Group: In this past year, I have also officially and non-officially mentored other SDEs. Thinking Big should not be limited to the individual. It's important to also foster it in others. As I grow as a mentor, I will work to inspire others to think big on their own; to see it as the natural and correct way to design. One example of where I am working on this already is in the transition of my build czar role to Swepna Doss. Swepna and I have been discussing ways to better automate the build czar tasks and through the developer pain of running the builds, Swepna is becoming more apt to seek better solutions for running the build.

**Manager Summary:**

Jonathan works hard and puts the customer first. He gets a lot done, and is a positive influence on those around him. He is correct that frugality is an area for improvement. He often tries to do too much, too well - sometimes tradeoffs need to be made.

"Jon has difficulty in multitasking – when he was mentoring an intern, sometimes he would spend too much time focusing on helping him instead of other work"

"Jonathan's passion to do the right thing at times can lead him to expend more time and energy on a task than it justifies. He would do well to improve on his "business judgment" and gauging the Return-on-Investment (ROI) on going the extra mile versus doing "good enough". "

"I suppose it's possible that he spends too much time on coming up with the Right Way to do things, rather than accepting that sometimes, the only way to get it out the door on time is to use a suboptimal solution (c.f. Otter CSC, possibly the image block). However, since he usually absorbs the extra work time required to implement the Right Way by working long hours, it doesn't have too much of an impact. "

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| <b>Ratings</b> |
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**Performance Rating:** Exceeds

**Leadership Principles Rating:** Role Model